

Dear Editor-in-Chief,

We submit "Innovation-Normalized Detection in Compound-Gaussian Clutter: Exact Laws, Conformal Thresholds and Certified Integration under Pulsed Interference" as a Regular Paper in IEEE Transactions on Radar Systems. The paper addresses slow-time CFAR detection and pulse integration under pulsed interference.

Under a true AR compound-Gaussian model with a common episode texture and no thermal noise, whitening and innovation normalization recover the classical CA and OS laws. We derive a post-onset detection law, its self-masking horizon and the delay provided by a slow-time guard. For exchangeable clean episodes, a conformal threshold on a worst-case bound certifies clipped and binary integration against arbitrary-amplitude interference under a dominating, clutter-independent hit model. These are conditional contributions; general empirical superiority is not claimed.

Measured validation uses IPIX and NetRAD sea clutter, JKU 77 GHz FMCW recordings and camera-labeled University of Washington radar recordings. On IPIX, conformal thresholds keep twelve of thirteen statistics within 0.73–1.47 times design down to 10^{-4} . On NetRAD, excluded from initial development, refitted thresholds stay within 1.36 times design at 10^{-3} ; five statistics exceed twice design at 10^{-4} . The per-look score shares the PAMF's whitening gain.

Three prospectively registered studies strengthen the evidence and delimit its scope. Injected known-onset trajectories through 48 looks confirm delayed, eventual collapse. The guarded screen acquires only 0.004–0.030 across all 24 idealized continuous-migration conditions; P-ANMF has a higher point estimate in every condition. All 3 camera-labeled recordings fail the correlation gate and provide no primary clean-entry units. Their uncertain depth and timing support descriptive presence association, with no demonstrated whitening advantage. Independent, precisely timed real-target onsets in correlated clutter remain unvalidated.

The supplement retains proofs, uncertainty, failures and protocol deviations, including burst-interference and sensitivity limits. Code, protocols and frozen numerical logs are available at <https://github.com/razaumair2203-ux/inarcp> (tag r22-trs-submission). AI assistance is disclosed in the Acknowledgment.

DRAFT — author confirmation required before submission: confirm originality and any relationship to prior publications, absence of concurrent review, any previous rejection by any journal and the required correspondence, and approval by every author. Complete author ORCIDs, funding and competing-interest declarations in the submission package. These facts cannot be certified from the local research artifacts.

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